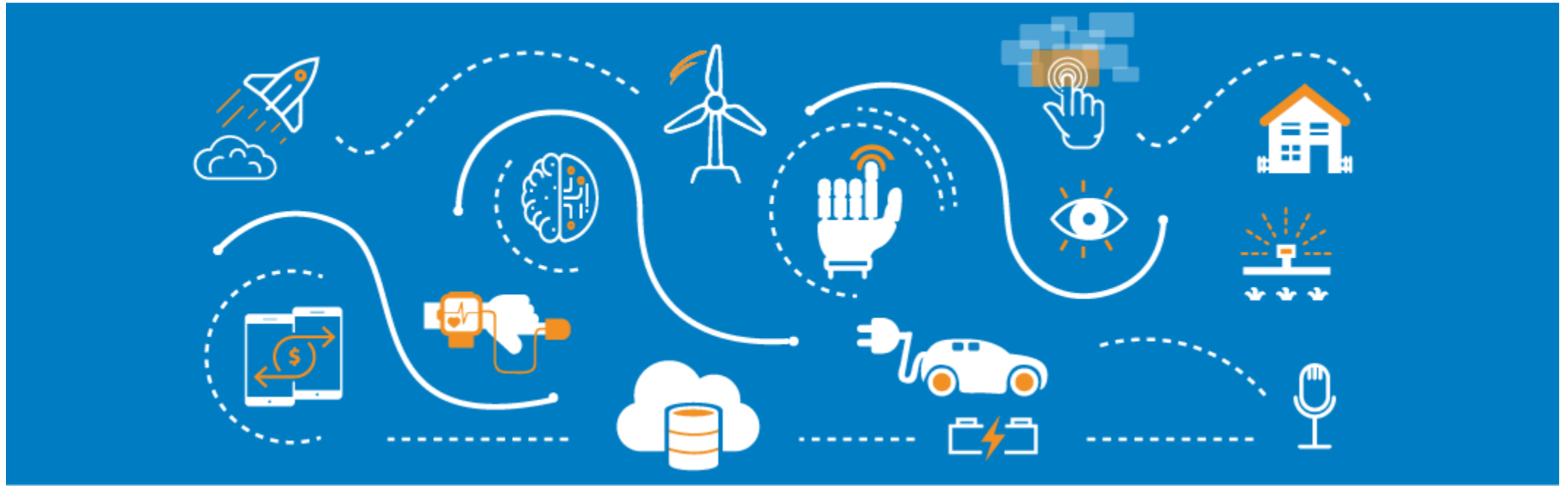




Computer Vision Part 4

Kien Tran

Contents

- 1** Dataset format and Dataset Preparation
- 2** Training SSD Model with Custom Data for Object Detection
- 3** Training YOLOv7 with Custom Data for Object Detection
- 4** Training YOLOv8 with Custom Data for Object Detection
- 5** Training YOLOv9 with Custom Data for Object Detection
- 6** Project Lab: MLOps

1. Dataset format and Dataset Preparation

1.1. Introduction

Pascal VOC and MS COCO are two most popular data format for object detection.

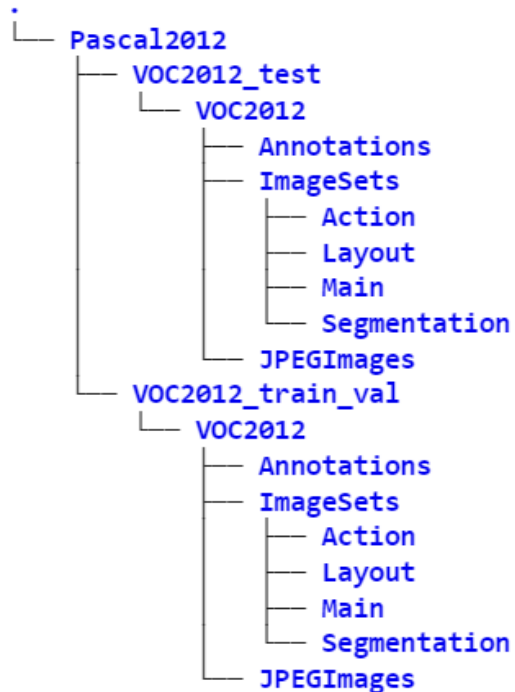
Most public available object detection datasets follow either one of these two formats.

- The Pascal Visual Object Classes ([VOC](#)) dataset for object detection and segmentation with 20 object classes and over 11K images.
- [COCO](#): Common Objects in Context (COCO) is a large-scale object detection, segmentation, and captioning dataset with 80 object categories. (Over 200'000 images of the total 330'000 images are labeled and 1.5 Mio object instances)

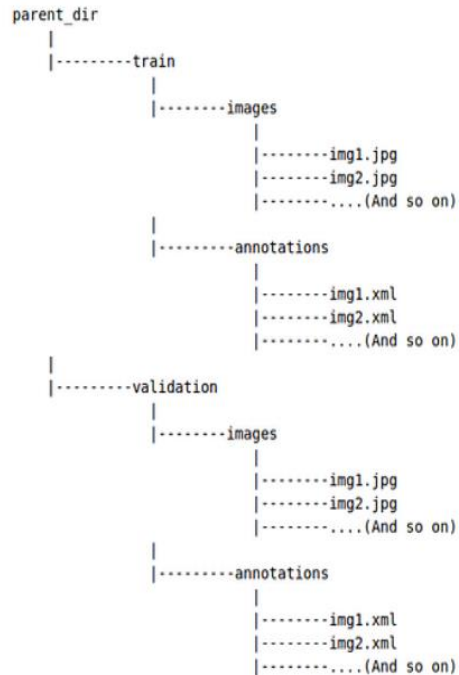
1. Data Format For Object Detection

1.2. Popular data format

```
!tree -d
```



Dataset required - in Pascal VOC format



```
<annotation>
  <folder>Kangaroo</folder>
  <filename>00001.jpg</filename>
  <path>./Kangaroo/stock-12.jpg</path>
  <source>
    <database>Kangaroo</database>
  </source>
  <size>
    <width>450</width>
    <height>319</height>
    <depth>3</depth>
  </size>
  <segmented>0</segmented>
  <object>
    <name>kangaroo</name>
    <pose>Unspecified</pose>
    <truncated>0</truncated>
    <difficult>0</difficult>
    <bndbox>
      <xmin>233</xmin>
      <ymin>89</ymin>
      <xmax>386</xmax>
      <ymax>262</ymax>
    </bndbox>
  </object>
</annotation>
```

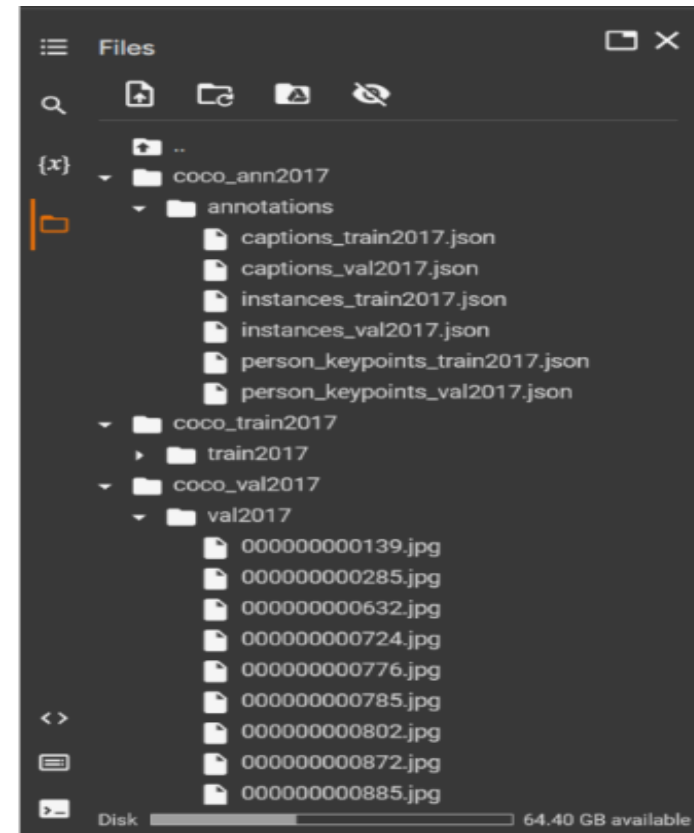
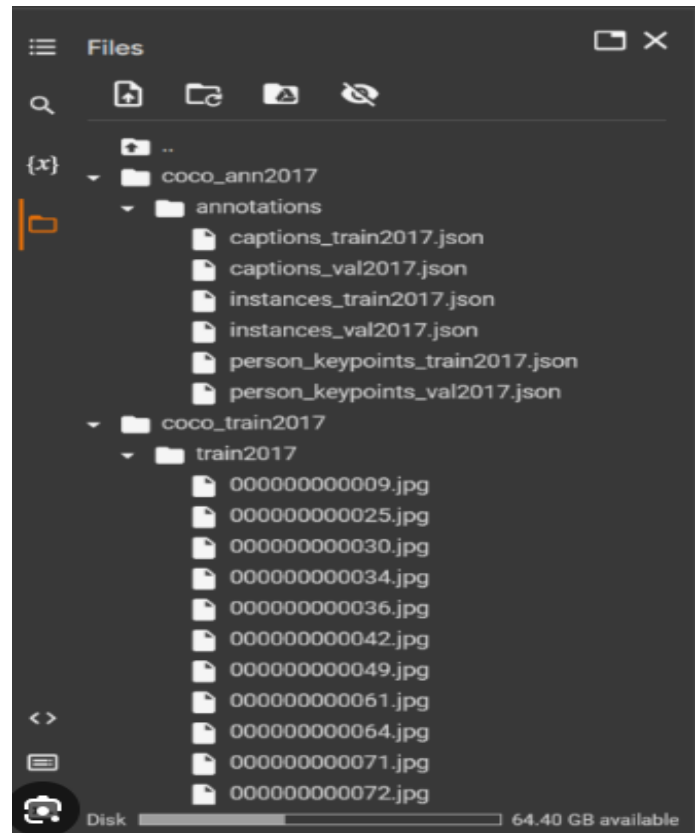
VOC Format (XML file)

Pascal VOC Bounding box :(*xmin-top left, ymin-top left,xmax-bottom right, ymax-bottom right*)

Sample Pascal VOC

1. Data Format For Object Detection

1.2. Popular data format



[COCO Format \(JSON file\)](#)

COCO Bounding box: (*x-top left, y-top left, width, height*)

1. Data Format For Object Detection

1.2. Popular data format

```
{
  "info": info,
  "licenses": [license],
  "categories": [category],
  "images": [image],
  "annotations": [annotation]
}
```

Categories:

```
Categories
[
  {
    "id": int,
    "name": str,
    "supercategory": str,
  }
]

"categories": [
  {
    "id": 1, "name": "rose", "supercategory": "flower"},
    {"id": 2, "name": "tulip", "supercategory": "flower"},
    {"id": 10, "name": "Apple", "supercategory": "fruit"}
  ]
]
```

template and example for Categories section of the JSON for COCO

Image: Contains list of all the images in dataset

```
image{
  "id": int,
  "width": int,
  "height": int,
  "file_name": str,
  "license": int,
  "flickr_url": str,
  "coco_url": str,
  "date_captured": datetime,
}

"images": [
  {
    "id": 397133,
    "width": 640,
    "height": 640,
    "file_name": "101.jpg",
    "license": 1,
    "date_captured": "2019-12-04 17:02:52"
  },
  {
    "id": 397122,
    "width": 640,
    "height": 640,
    "file_name": "102.jpg",
    "license": 1,
    "date_captured": "2019-12-04 17:02:52"
  }
]
```

template and example for images section of the json for COCO

Annoations: Contains list of each individual object annotation from every single image in the dataset

```
annotation{
  "id": int,
  "image_id": int,
  "category_id": int,
  "segmentation": RLE or [polygon],
  "area": float,
  "bbox": [x.y.width.height],
  "iscrowd": 0 or 1,
}

"annotations": [
  {
    "segmentation": [[510.66,423.01,511.72,420.03,...,510.45,423.01]],
    "area": 702.10,
    "iscrowd": 0,
    "image_id": 397133,
    "bbox": [433.07,355.93,138.65,228.67],
    "category_id": 18,
    "id": 1768
  },
  {
    "segmentation": {
      "counts": [12,56,198,10]
      "size": [120, 240]
    }
    "area": 500.2,
    "iscrowd": 1,
    "image_id": 397122,
    "bbox": [473.07,395.93,38.65,28.67],
    "category_id": 18,
    "id": 1768
  }
]
```

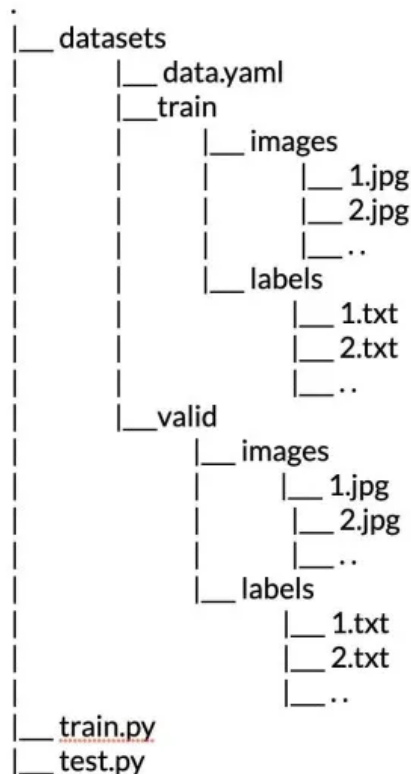
template and example for annotations section of the JSON for COCO

COCO Format (JSON file)

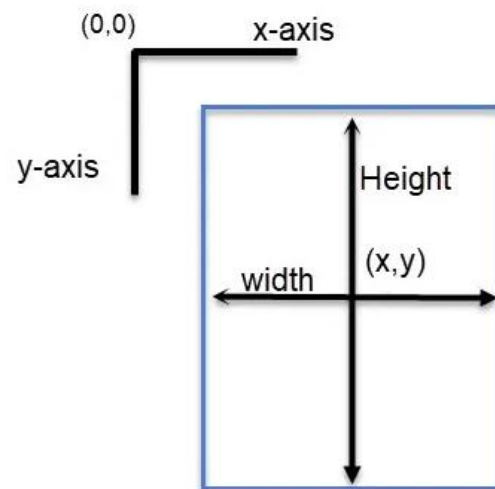
COCO Bounding box: (*x-top left, y-top left, width, height*)

1. Data Format For Object Detection

1.2. Popular data format



Sample folder structure for training using Ultralytics package. (Source: Image by the author.)



Bounding box format

```
names:
- bus
- car
- motorcycle
- truck
nc: 4
roboflow:
  license: CC BY 4.0
  project: drug_bottle_detection
  url: https://universe.roboflow.com/tma-v5lto/hcm-data/dataset/1
  version: 1
  workspace: tma-v5lto
test: ../test/images
train: ../train/images
val: ../valid/images
```

File config: data.yaml

Yolo Format (txt file)

COCO Bounding box: (*x center, y center, width, height*)

1. Data Format For Object Detection

1.3. Prepare Data with Roboflow

Roboflow Universe > TMA > HCM data

☆ HCM data Computer Vision Project

[Download this Dataset](#)

Images

2470 images [Explore Dataset](#)

Cite This Project

If you use this dataset in a research paper, please cite it using the following BibTeX:

```
@misc{
  hcm-data_dataset,
  title = { HCM data Dataset },
  type = { Open Source Dataset },
  author = { TMA },
  howpublished = { \url{ https://universe.roboflow.com/tma-v5lto/hcm-data } },
  url = { https://universe.roboflow.com/tma-v5lto/hcm-data },
  journal = { Roboflow Universe },
  publisher = { Roboflow },
  year = { 2024 },
  month = { may },
  note = { visited on 2024-05-26 },
}
```

SOURCE
TMA

LAST UPDATED
33 Minutes Ago

PROJECT TYPE
Object Detection

SUBJECT
vehicle

VIIEWS: 0

DOWNLOADS: 0

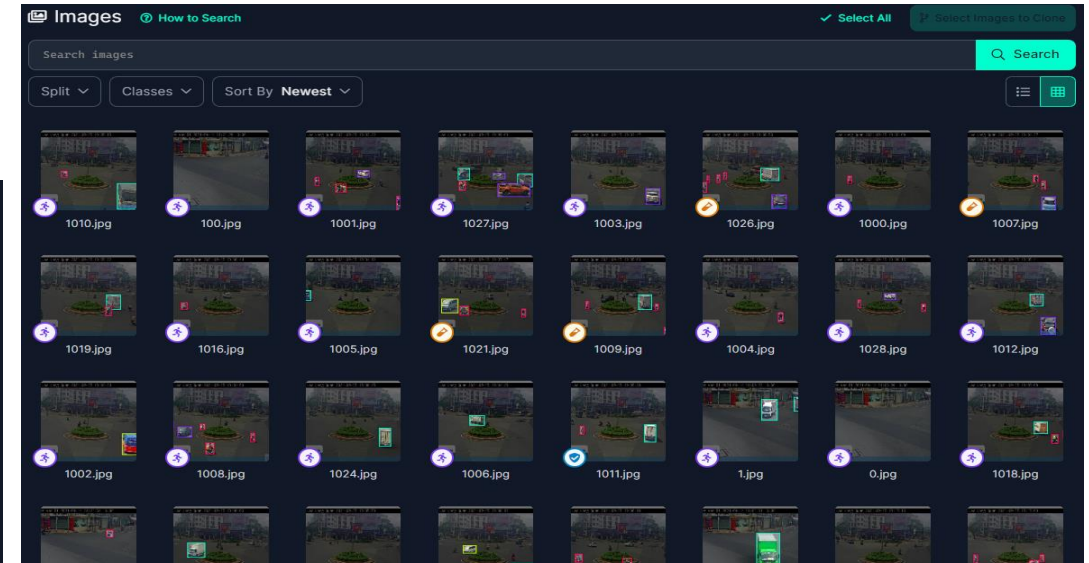
LICENSE
CC BY 4.0

CLASSES
bus car motorcycle truck

Object Detection

<https://roboflow.com>

<https://universe.roboflow.com/tma-v5lto/hcm-data/dataset/1>



HCM data Image Dataset

VERSIONS

v1 2024-05-26 4:00pm
Generated on May 26, 2024 [Download Dataset](#)

Popular Download Formats

YOLOv9 YOLOv8 YOLOv5 YOLOv7

COCO JSON YOLO Darknet Pascal VOC XML TFRecord

PaliGemma CreateML JSON Other Formats

2470 Total Images [View All Images](#)

Dataset Split

TRAIN SET 75% 1845 Images	VALID SET 15% 377 Images	TEST SET 10% 248 Images
-------------------------------------	------------------------------------	-----------------------------------

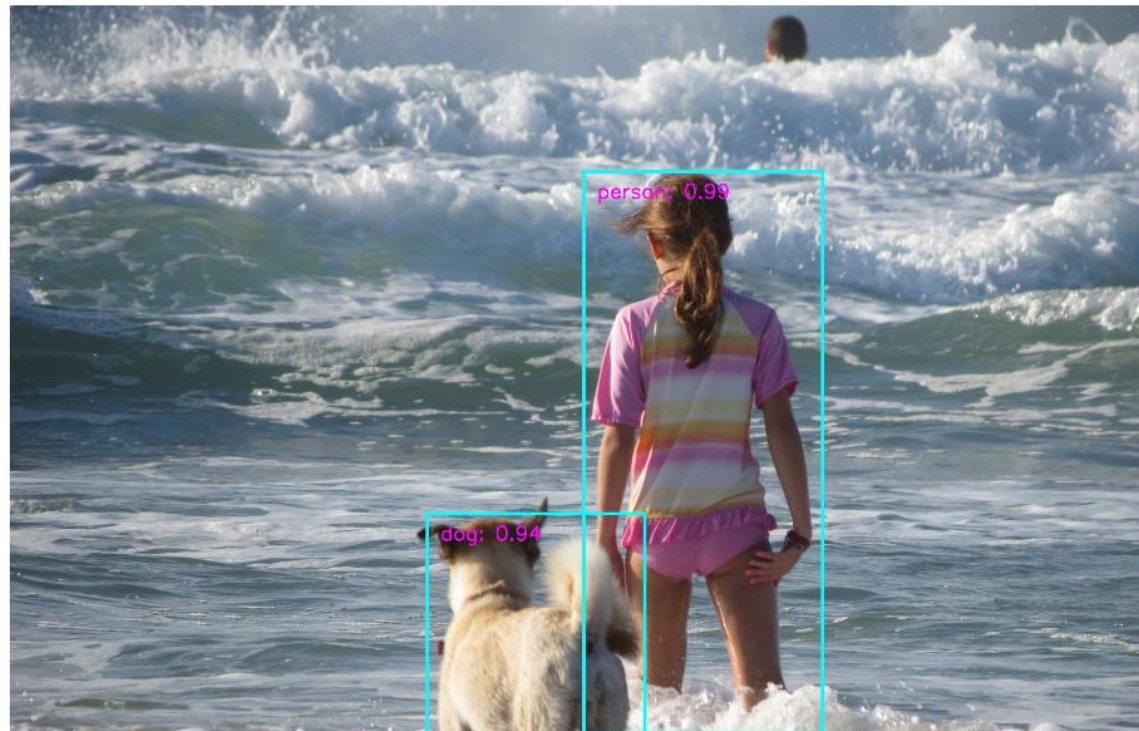
2. Training SSD Model with Custom Data

Single Shot MultiBox Detector Implementation in Pytorch

This repo implements [SSD \(Single Shot MultiBox Detector\)](#). The implementation is heavily influenced by the projects [ssd.pytorch](#) and [Detectron](#). The design goal is modularity and extensibility.

Currently, it has MobileNetV1, MobileNetV2, and VGG based SSD/SSD-Lite implementations.

It also has out-of-box support for retraining on Google Open Images dataset.

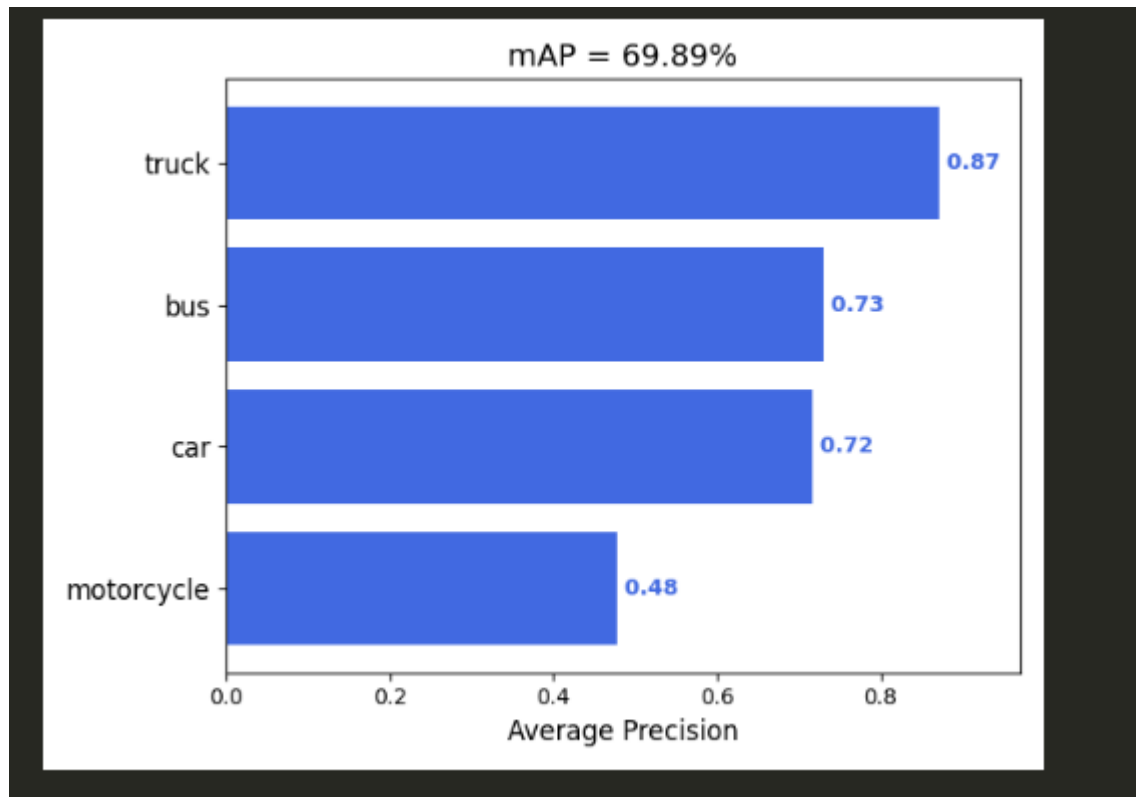


Ref: <https://colab.research.google.com/drive/1acFO-dSF9Wo4m0YqgCeHma17NiwOIKuV?usp=sharing>

Ref: <https://github.com/qfgaohao/pytorch-ssd/tree/master>

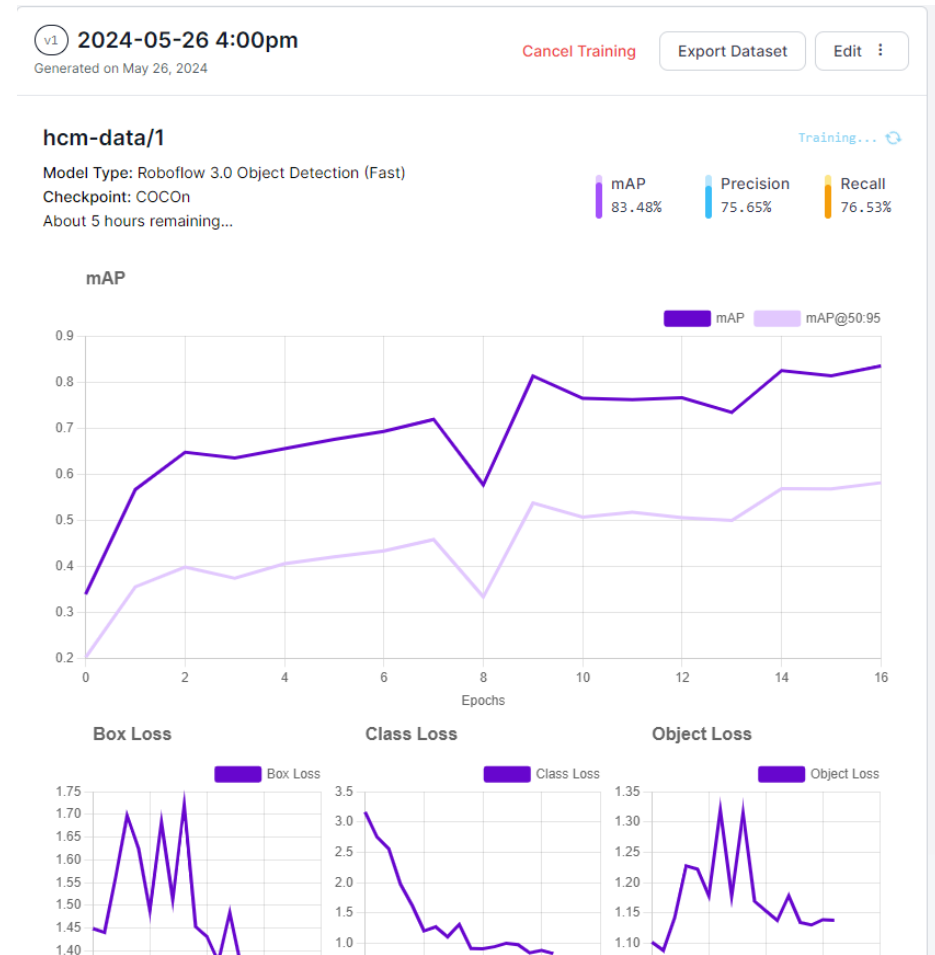
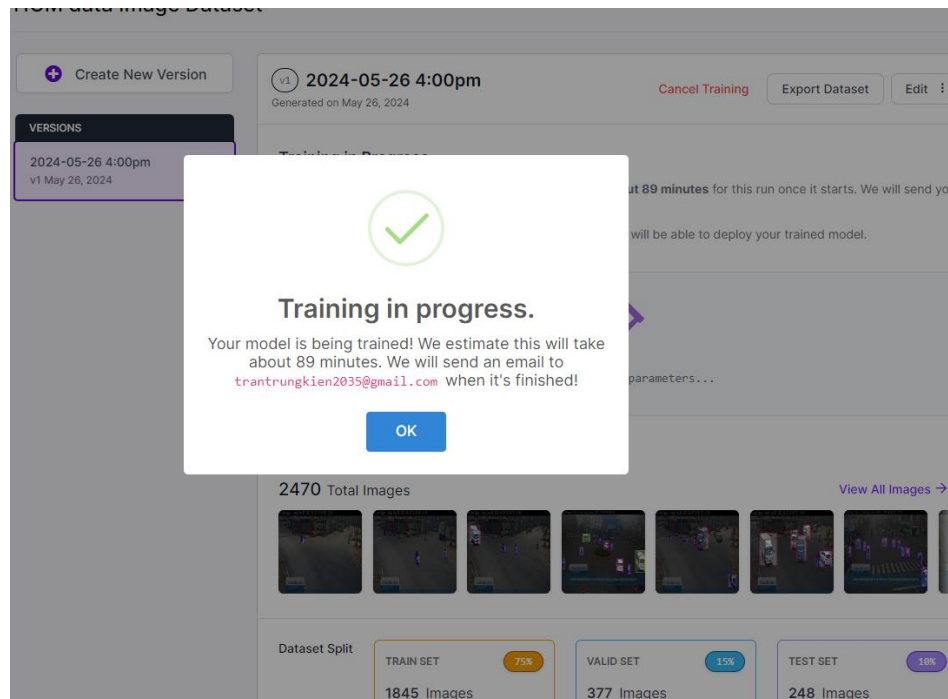
2. Training SSD Model with Custom Data

```
2021-12-07 02:40:30,608 - root - INFO - Epoch: 0, Validation Loss: 3.6739, Validation Regression Loss 1.1796, Validation Classification Loss: 2.4943
2021-12-07 02:40:30,799 - root - INFO - Saved model models/vgg16-ssd-Epoch-0-Loss-3.6738986587524414.pth
2021-12-07 02:56:42,228 - root - INFO - Epoch: 5, Validation Loss: 2.6860, Validation Regression Loss 0.7542, Validation Classification Loss: 1.9318
2021-12-07 02:56:42,394 - root - INFO - Saved model models/vgg16-ssd-Epoch-5-Loss-2.6860147094726563.pth
2021-12-07 03:12:50,147 - root - INFO - Epoch: 10, Validation Loss: 2.4596, Validation Regression Loss 0.6233, Validation Classification Loss: 1.8363
2021-12-07 03:12:50,312 - root - INFO - Saved model models/vgg16-ssd-Epoch-10-Loss-2.459552125930786.pth
2021-12-07 03:28:56,121 - root - INFO - Epoch: 15, Validation Loss: 2.3499, Validation Regression Loss 0.5893, Validation Classification Loss: 1.7606
2021-12-07 03:28:56,286 - root - INFO - Saved model models/vgg16-ssd-Epoch-15-Loss-2.349937158862540.pth
```



3. Training Yolov7 Model with Custom Data

Training with Roboflow

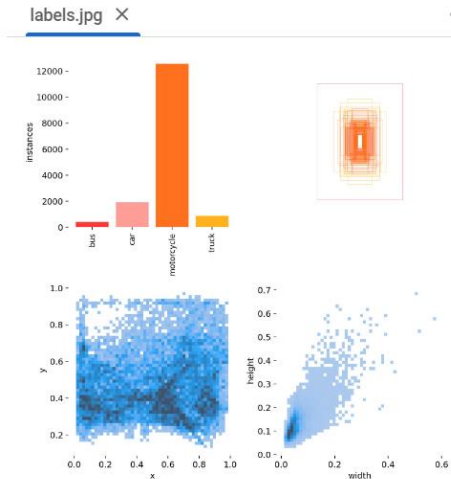


4. Training Yolov8 Model with Custom Data

... Downloading <https://ultralytics.com/assets/Arial.ttf> to '/root/.config/Ultralytics/Arial.ttf'...
100% 755k/755k [00:00<00:00, 12.8MB/s]
Overriding model.yaml nc=80 with nc=4

		from	n	params	module	arguments
0		-1	1	1392	ultralytics.nn.modules.conv.Conv	[3, 48, 3, 2]
1		-1	1	41664	ultralytics.nn.modules.conv.Conv	[48, 96, 3, 2]
2		-1	2	111360	ultralytics.nn.modules.block.C2f	[96, 96, 2, True]
3		-1	1	166272	ultralytics.nn.modules.conv.Conv	[96, 192, 3, 2]
4		-1	4	813312	ultralytics.nn.modules.block.C2f	[192, 192, 4, True]
5		-1	1	664320	ultralytics.nn.modules.conv.Conv	[192, 384, 3, 2]
6		-1	4	3248640	ultralytics.nn.modules.block.C2f	[384, 384, 4, True]
7		-1	1	1991808	ultralytics.nn.modules.conv.Conv	[384, 576, 3, 2]
8		-1	2	3985920	ultralytics.nn.modules.block.C2f	[576, 576, 2, True]
9		-1	1	831168	ultralytics.nn.modules.block.SPPF	[576, 576, 5]
10		-1	1	0	torch.nn.modules.upsampling.Upsample	[None, 2, 'nearest']
11		[-1, 6]	1	0	ultralytics.nn.modules.conv.Concat	[1]
12		-1	2	1993728	ultralytics.nn.modules.block.C2f	[960, 384, 2]
13		-1	1	0	torch.nn.modules.upsampling.Upsample	[None, 2, 'nearest']
14		[-1, 4]	1	0	ultralytics.nn.modules.conv.Concat	[1]
15		-1	2	517632	ultralytics.nn.modules.block.C2f	[576, 192, 2]
16		-1	1	332160	ultralytics.nn.modules.conv.Conv	[192, 192, 3, 2]
17		[-1, 12]	1	0	ultralytics.nn.modules.conv.Concat	[1]
18		-1	2	1846272	ultralytics.nn.modules.block.C2f	[576, 384, 2]
19		-1	1	1327872	ultralytics.nn.modules.conv.Conv	[384, 384, 3, 2]
20		[-1, 9]	1	0	ultralytics.nn.modules.conv.Concat	[1]
21		-1	2	4207104	ultralytics.nn.modules.block.C2f	[960, 576, 2]
22		[15, 18, 21]	1	3778012	ultralytics.nn.modules.head.Detect	[4, [192, 384, 576]]

Model summary: 295 layers, 25858636 parameters, 25858620 gradients, 79.1 GFLOPs



4. Training YOLOv8 Model with Custom Data

all	377	3021	0.815	0.85	0.878	0.639
-----	-----	------	-------	------	-------	-------

EarlyStopping: Training stopped early as no improvement observed in last 30 epochs. Best results observed
To update EarlyStopping(patience=30) pass a new patience value, i.e. `patience=300` or use `patience=0` to

```
NOTEBOOK
101 epochs completed in 2.072 hours.
Optimizer stripped from runs/detect/train/weights/last.pt, 52.0MB
Optimizer stripped from runs/detect/train/weights/best.pt, 52.0MB

Validating runs/detect/train/weights/best.pt...
Ultralytics YOLOv8.2.22 Python-3.10.12 torch-2.3.0+cu121 CUDA:0 (Tesla T4, 15102MiB)
Model summary (fused): 218 layers, 25842076 parameters, 0 gradients, 78.7 GFLOPs

```

Class	Images	Instances	Box(P	R	mAP50	mAP50-95)
all	377	3021	0.829	0.843	0.887	0.648
bus	377	81	0.834	0.827	0.828	0.663
car	377	391	0.828	0.859	0.914	0.686
motorcycle	377	2372	0.817	0.829	0.896	0.538
truck	377	177	0.835	0.859	0.911	0.704

```
Speed: 0.3ms preprocess, 11.1ms inference, 0.0ms loss, 6.3ms postprocess per image
Results saved to runs/detect/train
💡 Learn more at https://docs.ultralytics.com/modes/train
```


5. Training YOLOv9 v10 Model with Custom Data

Yolov9

Training

```
[ ] !yolo task=detect \
    mode=train \
    model=/content/drive/MyDrive/Colab/training_hcm/yolov9c.pt \
    data=/content/drive/MyDrive/Colab/training_hcm/hcm_data_yolov8/yolov7_9_data.yaml \
    patience=30 \
    batch=32 \
    epochs=150 \
    imgsz=640 \
    save=True \
    resume=True \
    project=/content/drive/MyDrive/Colab/training_hcm/hcm_data_yolov8/result_train9 \
    plots=True \
    device=0
```

Ultralytics YOLOv8.2.22 Python-3.10.12 torch-2.3.0+cu121 CUDA:0 (Tesla T4, 15102MiB)

Ref: <https://drive.google.com/file/d/1dHkc3iM-XYXhhzB9wgLihUhr6n42OTmc/view?usp=sharing>

Yolov10

Training

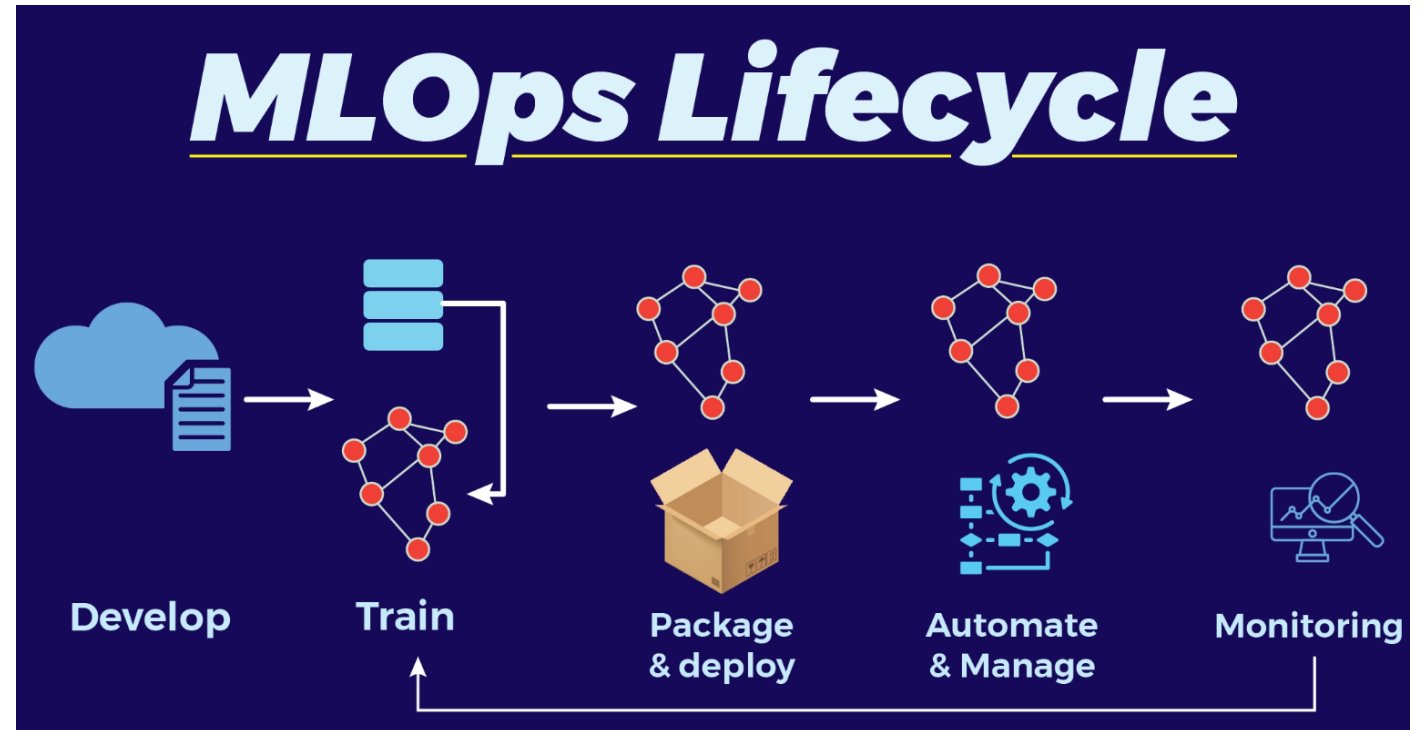
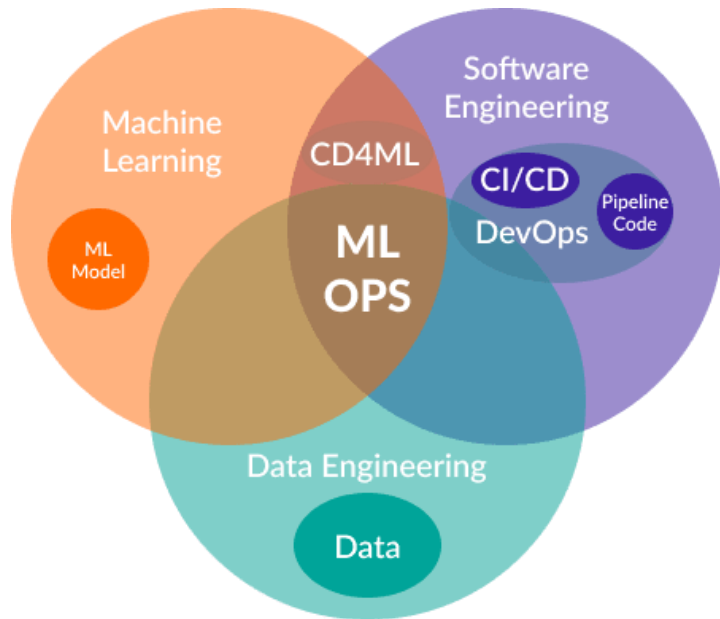
```
[ ] !yolo task=detect \
    mode=train \
    model=/content/drive/MyDrive/Colab/training_hcm/yolov10m.pt \
    data=/content/drive/MyDrive/Colab/training_hcm/hcm_data_yolov8/yolov7_9_data.yaml \
    patience=30 \
    batch=32 \
    epochs=150 \
    imgsz=640 \
    save=True \
    resume=True \
    project=/content/drive/MyDrive/Colab/training_hcm/hcm_data_yolov8/result_train10 \
    plots=True \
    device=0
```

Ultralytics YOLOv8.2.22 Python-3.10.12 torch-2.3.0+cu121 CUDA:0 (Tesla T4, 15102MiB)

Ref: <https://drive.google.com/file/d/1b1Qk3kqOizXZ5tnNMao8BwSU32hUpBA9/view?usp=sharing>

6. Project Lab: MLOps

6.0. MLOps



MLOps stands for Machine Learning Operations. MLOps is a core function of Machine Learning engineering, focused on streamlining the process of taking machine learning models to production, and then maintaining and monitoring them. MLOps is a collaborative function, often comprising data scientists, devops engineers, and IT.

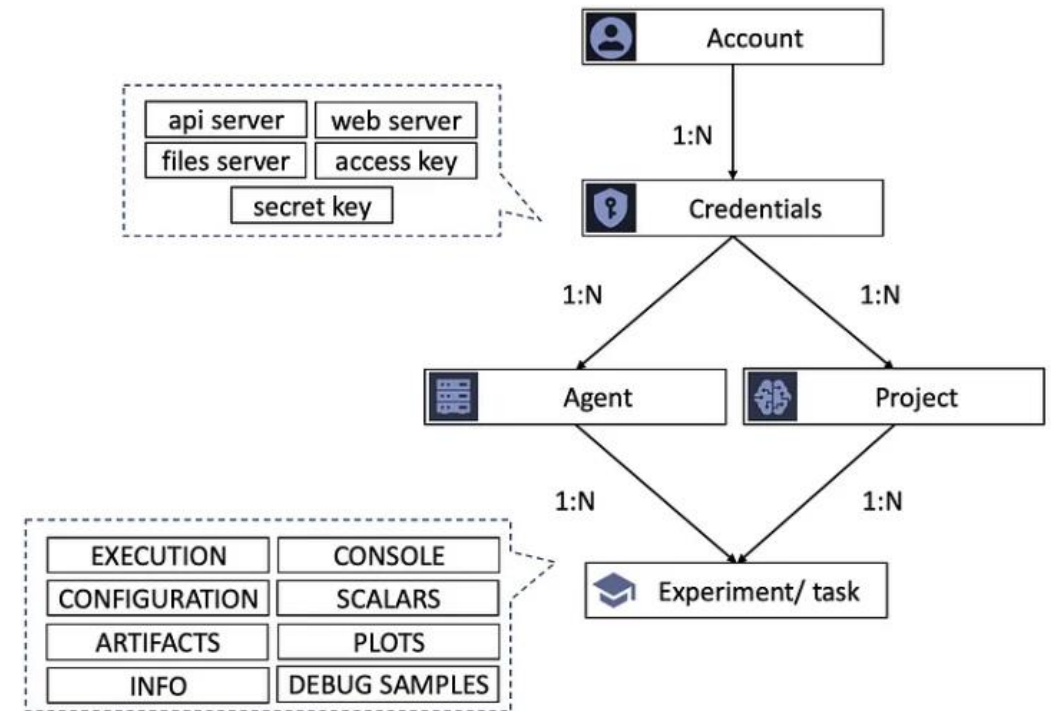
6. Project Lab: MLops

6.1. Basic concept

ClearML is an open source platform that automates and simplifies developing and managing machine learning solutions.

Key features:

- Real-time metrics tracking (loss, accuracy, validation)
- Experiment comparison
- Detailed logs and outputs
- Resource utilization monitoring
- Model Artifacts Management



Structure of ClearML

6. Project Lab: MLops

6.2. Set up

```
!pip install clearml
from clearml import Task, Logger
```

```
# Method 1) Set the environmental variables

%env CLEARML_WEB_HOST=https://app.clear.ml
%env CLEARML_API_HOST=https://api.clear.ml
%env CLEARML_FILES_HOST=https://files.clear.ml
%env CLEARML_API_ACCESS_KEY= <your access key>
%env CLEARML_API_SECRET_KEY= <your secret key>

# Method 2) Specify in Task object

Task.set_credentials(
    api_host="https://api.clear.ml",
    web_host="https://app.clear.ml",
    files_host="https://files.clear.ml",
    key='<your access key>',
    secret='<your secret key>'
)
```

6. Project Lab: MLops

6.3. Monitoring

The screenshot displays the Ultralytics ClearML interface for monitoring ML projects. The top navigation bar includes 'OVERVIEW', 'EXPERIMENTS' (highlighted), and 'MODELS'. The left sidebar shows a list of experiments under 'Kien Tran Trung's workspace / PROJECTS / my_project'. Three experiments are listed: 'my_yolov8_task2' (Running, updated 3 minutes ago), 'my_yolov8_task1' (Running, updated 6 minutes ago), and 'my_yolov8_task' (Draft, updated 12 minutes ago). The main panel shows the details for 'my_yolov8_task1' in the 'RUNNING' state. The 'CONSOLE' tab is active, displaying a log of training progress. The log includes a model summary, layer details, and training status updates.

my_yolov8_task1 (ID: a861e31b...)

EXECUTION CONFIGURATION ARTIFACTS INFO **CONSOLE** SCALARS PLOTS DEBUG SAMPLES

Hostname: cisai-MS-7D88

Download full log Filter: Filter By Regex

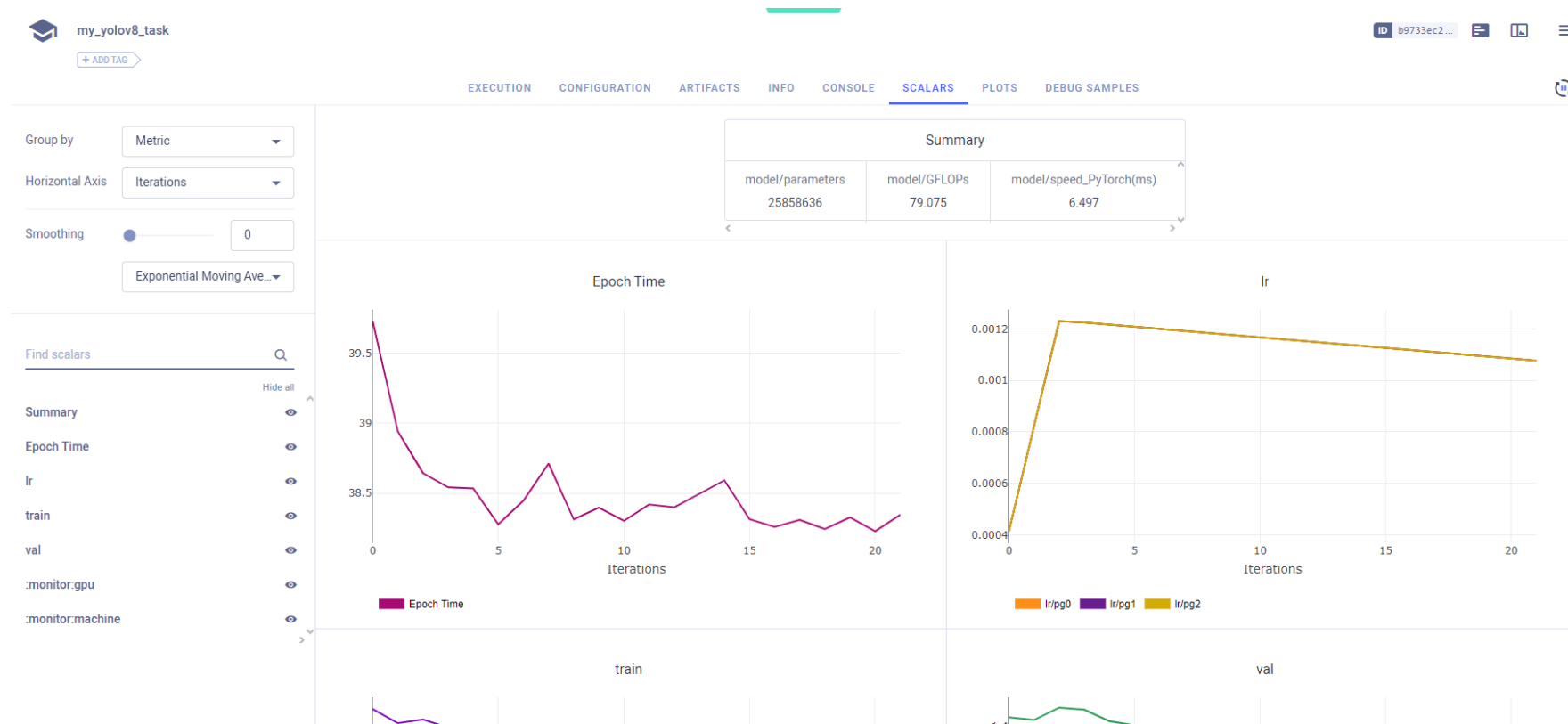
```
16      -1 1 332160 ultralytics.nn.modules.conv.Conv [192, 192, 3, 2]
17      [-1, 12] 1 0 ultralytics.nn.modules.conv.Concat [1]
18      -1 2 1846272 ultralytics.nn.modules.block.C2f [576, 384, 2]
19      -1 1 1327872 ultralytics.nn.modules.conv.Conv [384, 384, 3, 2]
20      [-1, 9] 1 0 ultralytics.nn.modules.conv.Concat [1]
21      -1 2 4207104 ultralytics.nn.modules.block.C2f [960, 576, 2]
22      [15, 18, 21] 1 3778012 ultralytics.nn.modules.head.Detect [4, [192, 384, 576]]

Model summary: 295 layers, 25858636 parameters, 25858620 gradients, 79.1 GFLOPs
Transferred 469/475 items from pretrained weights
2024-05-27 00:18:53 Freezing layer 'model.22.dfl.conv.weight'
AMP: running Automatic Mixed Precision (AMP) checks with YOLOv8n...
2024-05-27 00:18:53 /home/cisai/miniconda3/envs/yolo/lib/python3.8/site-packages/torch/nn/modules/conv.py:456: UserWarning:
Plan failed with a cudnnException: CUDNN_BACKEND_EXECUTION_PLAN_DESCRIPTOR: cudnnFinalize Descriptor Failed cudnn_status: CUDNN_STATUS_NOT_SUPPORTED (Triggered in
2024-05-27 00:18:54 AMP: checks passed ✓
2024-05-27 00:18:54 train: Scanning /home/cisai/project/mlops/training_hcm/hcm_data_yolov8/train/labels.cache... 1845 images, 0 backgrounds,
val: Scanning /home/cisai/project/mlops/training_hcm/hcm_data_yolov8/valid/labels.cache... 377 images, 0 backgrounds, 0 c
2024-05-27 00:18:55 Plotting labels to runs/detect/train2/labels.jpg...
2024-05-27 00:18:57 optimizer: 'optimizer=auto' found, ignoring 'lr=0.01' and 'momentum=0.937' and determining best 'optimizer', 'lr' and 'momentum' automatically...
optimizer: AdamW(lr=0.00125, momentum=0.9) with parameter groups 77 weight(decay=0.0), 84 weight(decay=0.0005), 83 bias(decay=0.0)
Image sizes 640 train, 640 val
Using 8 dataloader workers
Logging results to runs/detect/train2
Starting training for 10 epochs...
Closing dataloader mosaic
```

Ref: <https://docs.ultralytics.com/integrations/clearml/#configuring-clearml>

6. Project Lab: MLops

6.3. Monitoring



Ref: <https://docs.ultralytics.com/integrations/clearml/#configuring-clearml>

THANK YOU!



Email: contact@tmainnovation.vn



Website: www.tmainnovation.vn

